daa22.py

```python

import heapq

n = int(input("Enter number of characters: "))
heap = []

for i in range(n):
    ch = input("Enter character: ")
    f = int(input("Enter frequency: "))
    heapq.heappush(heap, (f, i, ch))

while len(heap) > 1:
    f1, _, left = heapq.heappop(heap)
    f2, _, right = heapq.heappop(heap)
    heapq.heappush(heap, (f1 + f2, n, [left, right]))
    n += 1

def code(tree, s=""):
    if isinstance(tree, str):
        print(tree, ":", s)
    else:
        code(tree[0], s + "0")
        code(tree[1], s + "1")

print("Huffman Codes:")
if heap:
    code(heap[0][2])

```